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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

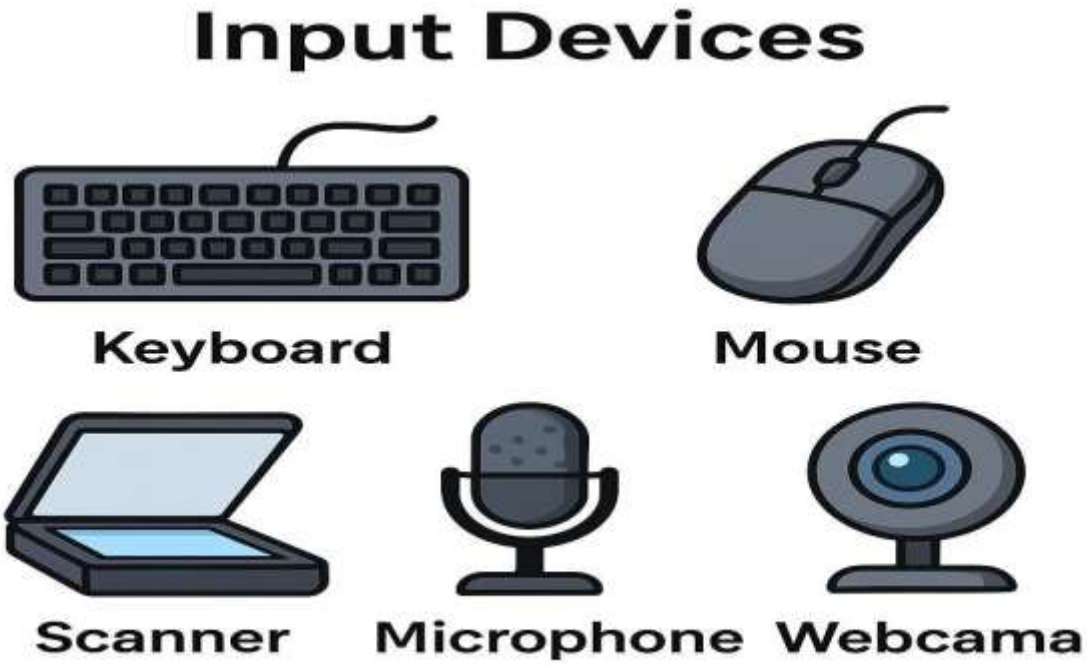
Daily Teaching : 2 Hours

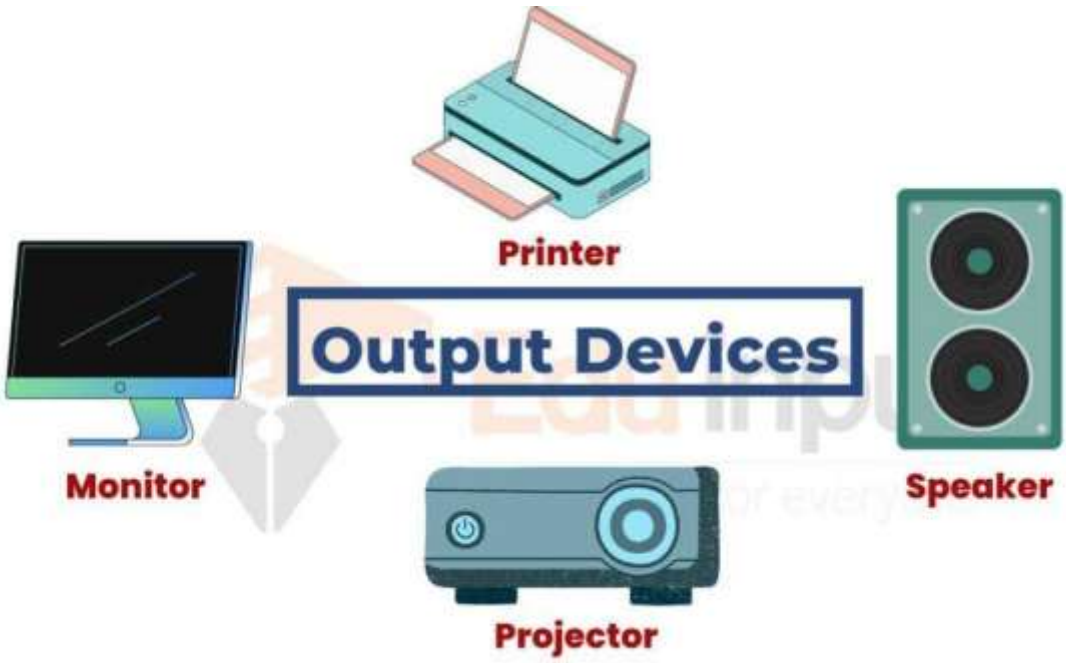
Total Classes : 100

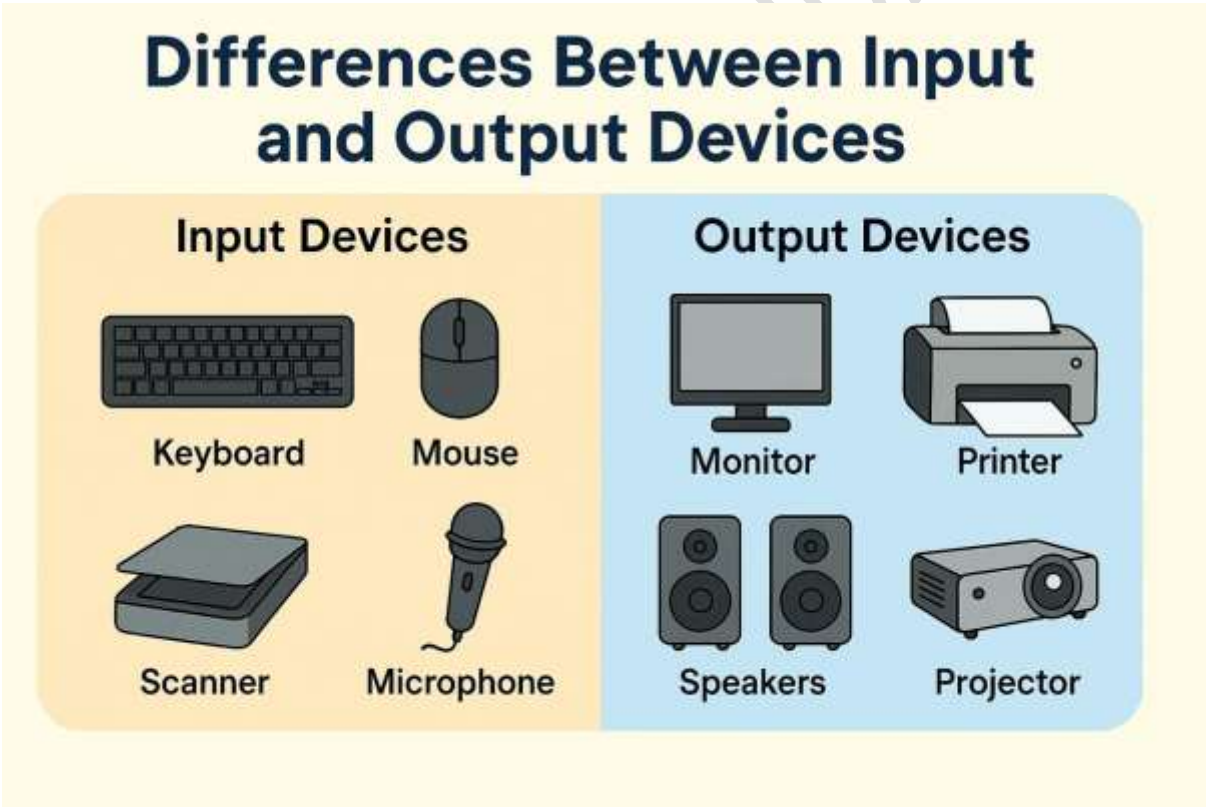
Level 2 – Young Technologist (7th–8th Standard)

Class 4: Input & Output Devices

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Learning Objectives

By the end of this lesson, students will be able to:

- Define **Input Devices** and **Output Devices**.

- Identify common input and output devices.
- Explain the functions of different input and output devices.
- Differentiate between input and output devices.
- Understand how data moves through a computer.
- Perform hands-on activities using input and output devices.

Introduction

A computer needs devices to **receive information** and **show results**.

- **Input Devices** help us enter data into a computer.
- **Output Devices** help the computer display or produce information for us.

Example

When you type your name:

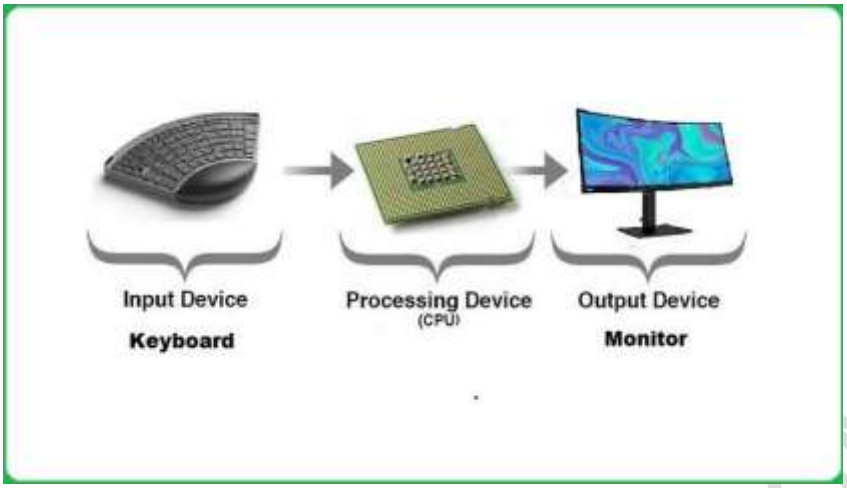
- ⌨️ Keyboard → Input Device
- 🧠 CPU → Processes the data
- 🖥️ Monitor → Output Device

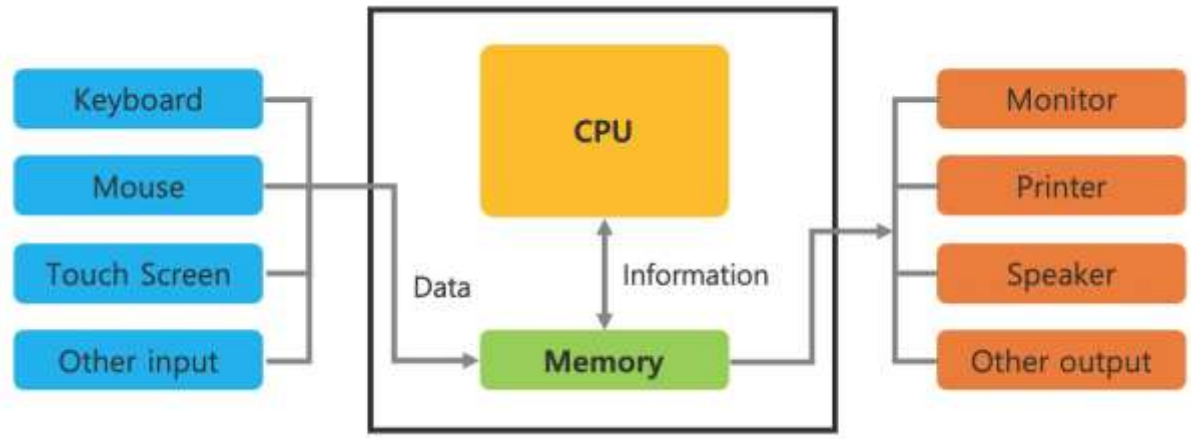
This process follows the **IPO Cycle (Input → Process → Output).**

IPO Cycle (Input – Process – Output)



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Step 1: Input

The user enters data using an input device.

⬇️

Step 2: Processing

The CPU processes the data.

⬇️

Step 3: Output

The result is shown using an output device.

Example

Step	Example
Input	Type **10 + 20** using the keyboard

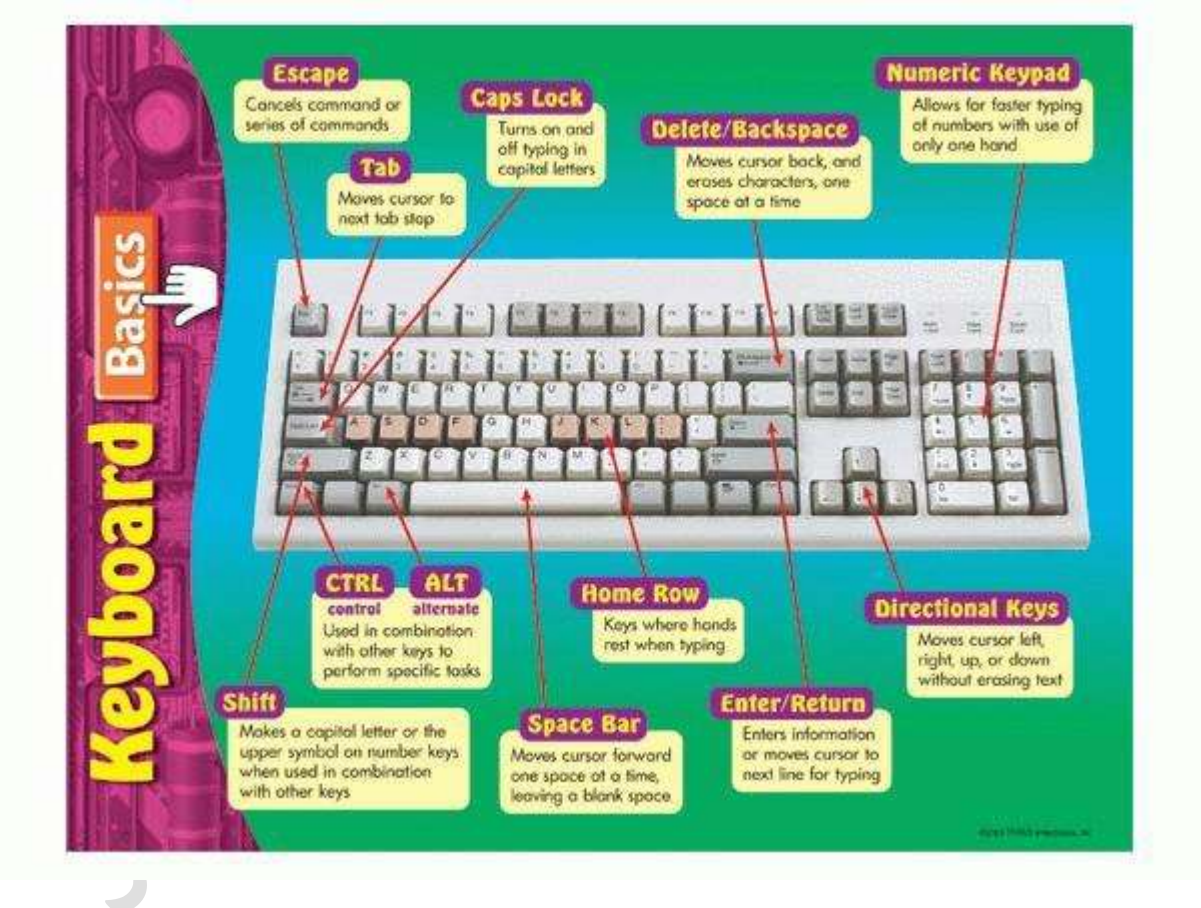
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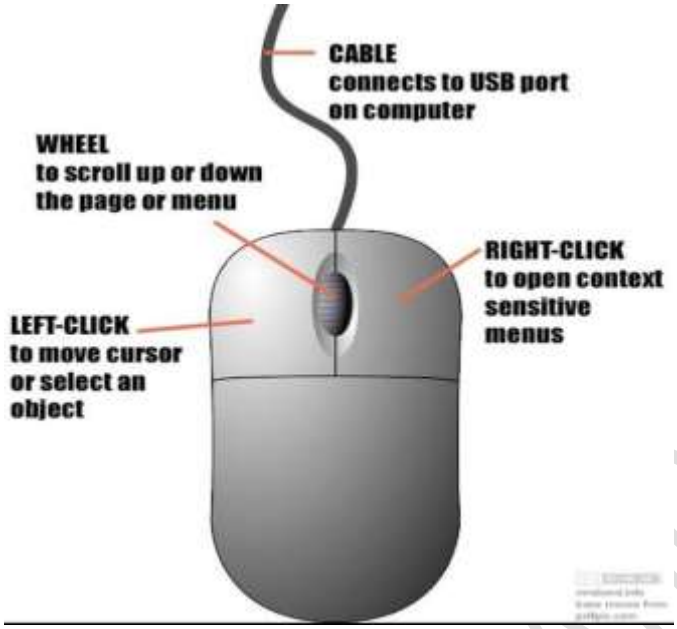
Step	Example
Processing	CPU calculates the answer
Output	Monitor displays **30**

What are Input Devices?

An **Input Device** is a hardware device used to **enter data and instructions** into a computer.

Common Input Devices







Scanner



OMR



Microphone



Joystick



Barcode Reader



Touch-Screen

1. Keyboard

The keyboard is the most common input device.

Uses

- Typing letters

- Typing numbers
- Writing documents
- Entering passwords
- Coding in Python

2. Mouse

The mouse helps us control the pointer on the screen.

Uses

- Click
- Double-click
- Right-click
- Drag and drop
- Select items

3. Scanner

A scanner converts paper documents or photos into digital files.

Uses

- Scan certificates
- Scan photographs
- Scan drawings

4. Microphone

A microphone records sound and voice.

Uses

- Online classes
- Voice recording
- Video calls

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- Voice search

5. Webcam

A webcam captures photos and videos.

Uses

- Online meetings
- Video calls
- Taking pictures

6. Joystick

A joystick is used mainly for games and simulations.

Uses

- Flying games
- Racing games
- Robotics control

7. Barcode Scanner

Reads barcodes printed on products.

Uses

- Supermarkets
- Shopping malls
- Libraries

8. Touchscreen

A touchscreen allows users to interact by touching the display.

Uses

- Smartphones

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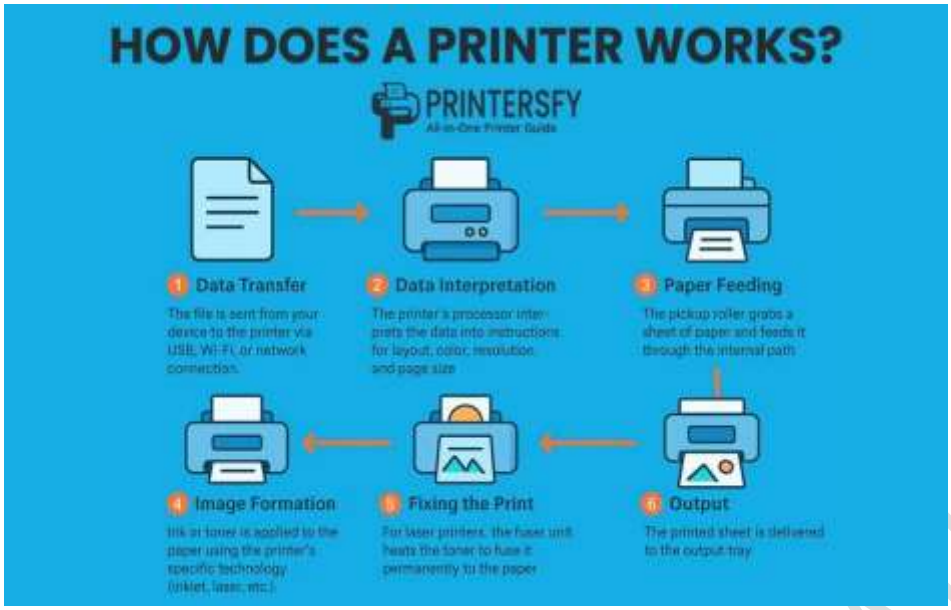
- Tablets
- ATM machines
- Self-service kiosks

What are Output Devices?

An **Output Device** is a hardware device that **shows or produces the results** processed by the computer.

Common Output Devices





1. Monitor

Displays text, pictures, videos, and games.

Uses

- Watching videos
- Reading documents
- Coding
- Playing educational games

2. Printer

Prints documents and images onto paper.

Types

- Inkjet Printer
- Laser Printer

3. Speakers

Produce sound from the computer.

Uses

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- Listening to music
- Watching videos
- Online classes

4. Headphones

Provide private audio without disturbing others.

5. Projector

Projects the computer screen onto a large wall or screen.

Uses

- Classrooms
- Conferences
- Movie presentations

6. Plotter

A plotter prints large drawings with high precision.

Uses

- Engineering
- Architecture
- Maps

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Difference Between Input and Output Devices

Input Devices	**Output Devices**
Enter data into the computer	Display or produce results
Used before processing	Used after processing
Controlled by the user	Controlled by the computer

Input Devices	Output Devices
Examples: Keyboard, Mouse, Scanner	Examples: Monitor, Printer, Speakers

Real-Life Examples

Activity	Input Device	Output Device
Typing homework	Keyboard	Monitor
Drawing in Paint	Mouse	Monitor
Printing a project	Keyboard, Mouse	Printer
Video call	Webcam, Microphone	Monitor, Speakers
Listening to music	Mouse	Speakers

How Input and Output Devices Work Together

Example 1 – Typing

1. Type using the **keyboard**.
2. CPU processes the letters.
3. **Monitor** displays the text.

Example 2 – Printing

1. Click **Print** using the mouse.
2. CPU processes the command.
3. **Printer** prints the document.

Example 3 – Online Class

1. Speak into the **microphone**.
2. CPU processes the voice.
3. **Speakers** play the teacher's voice.
4. **Monitor** displays the video.

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Advantages of Input Devices

- Easy data entry
- Fast communication
- Better interaction
- Supports creative work

Advantages of Output Devices

- Easy to understand information
- Clear display of results
- Helps in learning and presentations
- Produces printed documents

Computer Lab Safety

- Handle the keyboard and mouse gently.
- Do not pull cables.
- Keep devices clean.
- Do not touch screens with sharp objects.
- Use headphones at a safe volume.
- Turn devices off properly after use.

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Practical Activities

Practical Activity 1 – Identify Input & Output Devices

Objective

Identify the devices available in the computer lab.

Write their names and whether they are **Input** or **Output**.

Device	Type
Keyboard	Input
Mouse	Input
Monitor	Output
Printer	Output
Microphone	Input
Speakers	Output

Practical Activity 2 – Mouse Practice

Instructions

1. Open **Paint**.
2. Draw:
    - A house
    - A tree
    - The sun
3. Save the drawing.

Skills Learned

- Clicking
- Dragging
- Drawing
- Saving files

Practical Activity 3 – Keyboard Typing

Type the following:

My Name is _____.

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I am studying in Class ____.

I love learning computers.

Python is fun!

Artificial Intelligence is amazing.

Save the file as **MyIntroduction.docx** or **MyIntroduction.txt**.

Classroom Activity

Sort the Devices

Write **Input** or **Output**.

Device	**Answer**
Keyboard	Input
Monitor	Output
Scanner	Input
Printer	Output
Mouse	Input
Speakers	Output
Microphone	Input
Projector	Output
Touchscreen	Input (also acts as an output display)

Quick Quiz

1. Which device is used to type text?

A. Monitor

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B. Keyboard ✅

C. Printer

D. Speaker

2. Which device displays information?

A. Mouse

B. Monitor ✅

C. Scanner

D. Microphone

3. Which device records your voice?

A. Printer

B. Webcam

C. Microphone ✅

D. Speaker

4. Which device prints documents?

A. Printer ✅

B. Monitor

C. Mouse

D. Keyboard

5. Which device is mainly used for video calls?

A. Webcam ✅

B. Printer

C. Keyboard

D. Speaker

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Home Assignment

1. Draw and label **five input devices**.
2. Draw and label **five output devices**.
3. Write **three uses** of each:
    - Keyboard
    - Mouse
    - Monitor
    - Printer
4. Observe the computer devices at your home or school and classify them as **Input** or **Output**.

Lesson Summary

In this lesson, students learned:

- What input and output devices are.
- The IPO (Input–Process–Output) cycle.
- Common input devices such as the keyboard, mouse, scanner, microphone, webcam, joystick, barcode scanner, and touchscreen.
- Common output devices such as the monitor, printer, speakers, headphones, projector, and plotter.
- The differences between input and output devices.
- Safe handling of computer devices.
- Practical activities using the keyboard, mouse, and monitor.

Learning Outcomes

By the end of this lesson, students will be able to:

- Define input and output devices.
- Identify common input and output hardware.
- Explain the function of each device.

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- Describe how data flows through the IPO cycle.
- Use basic input devices like the keyboard and mouse confidently.
- Handle computer devices safely and responsibly.

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